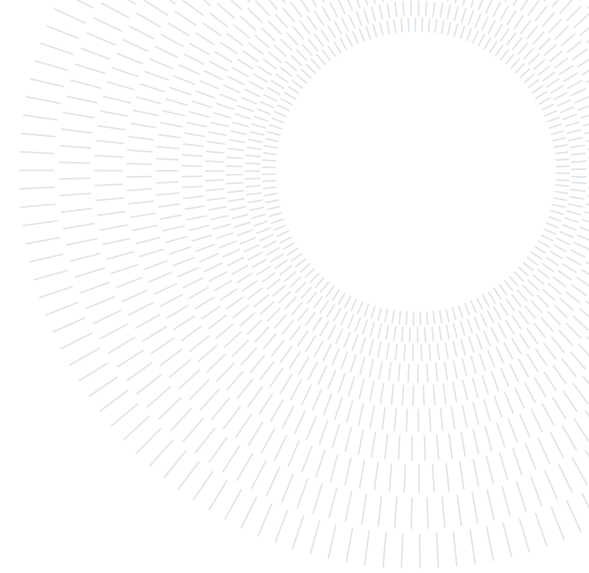




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NUMERICAL METHODS FOR PARTIAL DIFFERENTIAL EQUATIONS

Fisher-Kolmogorov Equation for Neurodegenerative Diseases

MASTER DEGREE IN HIGH PERFORMANCE COMPUTING ENGINEERING

Authors: ARIANNA AMADINI, FEDERICO QUARTIERI, ALESSANDRO RUZZA, DANIELE SALVI

Professor: PROF. ALFIO MARIA QUARTERONI

Teaching Assistant: DOTT. MICHELE BUCELLI

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1. Introduction

The Fisher-Kolmogorov equation, also known as the FK equation, is a fundamental reaction-diffusion model widely used in mathematical biology and physics. It describes the propagation of waves in systems where diffusion and reaction interact, making it a key tool for modeling various natural phenomena. One of its significant applications is in understanding the spread of misfolded proteins in neurodegenerative diseases such as Alzheimer's and Parkinson's. This equation helps capture how pathological proteins spread through the brain, contributing to disease progression.

Originally introduced to describe the spread of advantageous genetic traits in populations, the FK equation has been extensively studied in the context of traveling wave solutions. These solutions provide insights into wave propagation, stability, and speed, which are crucial for understanding real-world applications such as epidemiology, tumor growth, and chemical reactions.

In this work, we analyze the Fisher-Kolmogorov equation both numerically and theoretically. Using computational tools such as the Deal.II library, we explore the behavior of solutions under different initial conditions and study the stability of traveling waves. The goal is to understand how variations in diffusion and reaction parameters affect the spread dynamics in different scenarios.

2. Mathematical Model

The mathematical formulation we worked with is given by the Fisher-Kolmogorov equation, which is formulated as a reaction-diffusion equation.

In particular, given a generic $t \in (0, T]$, $T > 0$ and $\mathbf{x} \in \Omega \subset \mathbb{R}^d$, find $c = c(\mathbf{x}, t)$ such that:

$$\begin{cases} \frac{\partial c}{\partial t} - \nabla \cdot (D \nabla c) - \alpha c(1 - c) = 0, & \text{in } \Omega \times (0, T], \\ D \nabla c \cdot \mathbf{n} = 0, & \text{on } \partial\Omega \times (0, T], \\ c(t = 0) = c_0, & \text{in } \Omega. \end{cases} \quad (1)$$

In the previous system:

- $c(\mathbf{x}, t) \in [0, 1]$ represents the concentration of misfolded proteins. In particular, the steady state solutions are obtained for $c = 0$ and $c = 1$. In $c = 0$ the solution is unstable, while in $c = 1$ it is stable, meaning that the concentration will always be repelled from the benign state, $c = 0$, and attracted to the misfolded state, $c = 1$. [1]
- $\alpha = \alpha(\mathbf{x}) \in \mathbb{R}^+$ is the reaction rate, modeling the conversion from healthy to misfolded states.
- $D = D(\mathbf{x})$ is the diffusion tensor, governing the spatial propagation of misfolded proteins and it's formulated as follows:

$$D = d^{\text{ext}} I + d^{\text{axn}} \mathbf{n} \otimes \mathbf{n}, \quad d^{\text{axn}} \geq d^{\text{ext}}.$$

We can clearly see that D has two components: the extracellular diffusion $d^{\text{ext}} \in \mathbb{R}^+$ and the axonal transport $d^{\text{axn}} \in \mathbb{R}^+$. d^{ext} is associated with the isotropic diffusion of misfolded proteins through the extracellular space, while $d^{\text{axn}} \in \mathbb{R}^+$ with the anisotropic diffusion along the axonal fibers direction $\mathbf{n}$. [1]

It is necessary to assume the following regularity assumptions in order to solve (1), according to [2]:

- $\alpha \in L^\infty(\Omega)$;
- $D \in L^\infty(\Omega, \mathbb{R}^{d \times d})$, and $\exists d_0, D_0 > 0 \forall \xi \in \mathbb{R}^d : d_0 |\xi|^2 \leq \xi^\top D \xi \leq D_0 |\xi|^2 \quad \forall \xi \in \mathbb{R}^d$;
- $c_0 \in L^2(\Omega)$.

Variable	Meaning	Unit
$c(\mathbf{x}, t)$	Concentration	dimensionless
$\alpha(\mathbf{x})$	Reaction rate	$years^{-1}$
$D(\mathbf{x})$	Diffusion tensor	$mm^d/years$

Table 1: Unit of measurement for each variable of the FK equation for a d-dimensional mesh.

It is possible to observe that a forcing term is not present, thus the system is not affected by external forces. Moreover, there are no Dirichlet conditions, meaning that the system is isolated.

To analyze the behavior of the solutions, we explore the finite element method and investigate the influence of various parameters on wave speed and stability, providing insights into the fundamental mechanisms underlying neurodegenerative disease progression.

2.1. Weak Formulation

Since $\partial\Omega = \Gamma_N$, let $V = H^1(\Omega)$ and consider $v \in V$. We multiply by v and integrate over Ω the first equation of the strong formulation:

$$\int_{\Omega} \frac{\partial c}{\partial t} v \, dx - \int_{\Omega} \nabla \cdot (D \nabla c) v \, dx - \int_{\Omega} \alpha c(1 - c) v \, dx = 0. \quad (2)$$

Then, we integrate it by parts:

$$\int_{\Omega} \frac{\partial c}{\partial t} v \, dx + \int_{\Omega} D \nabla c \cdot \nabla v \, dx - \int_{\partial\Omega} D \nabla c \cdot \mathbf{n} v \, ds - \int_{\Omega} \alpha c(1 - c) v \, dx = 0. \quad (3)$$

We can simplify the term related to the Neumann boundary condition:

$$\int_{\partial\Omega} D \nabla c \cdot \mathbf{n} v \, ds = \int_{\partial\Omega} \phi_N v \, ds = 0. \quad (4)$$

So, with the weak formulation, $\forall t \in (0, T]$ we find $c(t) \in V$ such that:

$$\begin{cases} \int_{\Omega} \frac{\partial c}{\partial t} v \, dx + \int_{\Omega} D \nabla c \cdot \nabla v \, dx - \int_{\Omega} \alpha c(1 - c) v \, dx = 0 & \forall v \in V, \\ c(t = 0) = c_0. \end{cases} \quad (5)$$

Let us define:

- $a(c)(v) = \int_{\Omega} D \nabla c \cdot \nabla v \, dx - \int_{\Omega} \alpha c(1 - c) v \, dx$;
- $F(v) = 0$.

Now, we can define the residual as follows:

$$R(c)(v) = \int_{\Omega} \frac{\partial c}{\partial t} v \, dx + a(c)(v) - F(v). \quad (6)$$

Finally, the weak formulation can be rewritten by leveraging the definition of residual.

Find $c(t) \in V$ such that $\forall t \in (0, T]$:

$$\begin{cases} R(c)(v) = 0 & \forall v \in V, \\ c(t = 0) = c_0. \end{cases} \quad (7)$$

2.2. Semi Discrete Formulation

Let us now introduce a mesh on Ω , as will be explained in Section 3.1, and denote a triangulation of the domain Ω by:

$$\mathcal{T}_h = \{K\}, \quad \text{s.t.} \quad \overline{\Omega} = \bigcup_{K \in \mathcal{T}_h} K,$$

where each element $K \in \mathcal{T}_h$ is a triangle (in 2D) or a tetrahedron (in 3D) and:

$$h = \max_{K \in \mathcal{T}_h} \text{diam}(K), \quad \text{with} \quad \text{diam}(K) = \max_{x_i, x_j \in K} |x_i - x_j|.$$

We define $\mathbb{P}_r(K)$ as the space of polynomials of degree at most r on the mesh element K . Then, we introduce the following finite element space:

$$X_h^r = \{v_h : \Omega \rightarrow \mathbb{R} \mid v_h \in C^0(\overline{\Omega}), \quad v_h|_K \in \mathbb{P}^r(K) \quad \forall K \in \mathcal{T}_h\}, \quad r \geq 1.$$

Finally, we define $V_h = V \cap X_h^h(\Omega)$ in order to perform a spatial discretization.

The semi-discrete formulation can then be written as follows:

$\forall t \in (0, T)$, find $c_h(t) \in V_h$ such that:

$$\begin{cases} R(c_h)(v_h) = 0 & \forall v_h \in V_h, \\ c_h(t = 0) = c_{0,h}. \end{cases} \quad (8)$$

In other words, $\forall t \in (0, T)$, find $c_h(t) \in V_h$ such that:

$$\begin{cases} \int_{\Omega} \frac{\partial c_h}{\partial t} v_h \, dx + \int_{\Omega} D \nabla c_h \cdot \nabla v_h \, dx - \int_{\Omega} \alpha c_h (1 - c_h) v_h \, dx = 0 & \forall v_h \in V_h, \\ c_h(t = 0) = c_{0,h}. \end{cases} \quad (9)$$

2.3. Fully Discrete Formulation

We now introduce a temporal discretization of the interval $[0, T]$.

Firstly, let $0 = t_0 < t_1 < \dots < t_N = T$ be a uniform time grid with step Δt and let $V_h \subset H^1(\Omega)$ be the finite element space. We denote by $c_h^n \in V_h$ the approximation of $c(t_n)$.

Using the θ -method, the fully discrete scheme reads: for all $n = 0, 1, \dots, N-1$, find $c_h^{n+1} \in V_h$ such that:

$$\int_{\Omega} \frac{c_h^{n+1} - c_h^n}{\Delta t} v_h \, dx + \theta a(c_h^{n+1})(v_h) + (1 - \theta) a(c_h^n)(v_h) = 0, \quad \forall v_h \in V_h, \quad (10)$$

with initial condition $c_h^0 = c_{0,h} \in V_h$ and recalling from the weak formulation that:

$$a(c)(v) = \int_{\Omega} D \nabla c \cdot \nabla v \, dx - \int_{\Omega} \alpha c (1 - c) v \, dx.$$

Also, $\theta \in [0, 1]$ is a parameter controlling the time integration scheme and in particular:

- $\theta = 0$: Forward Euler method (explicit);
- $\theta = \frac{1}{2}$: Crank-Nicolson method (semi-implicit);
- $\theta = 1$: Backward Euler method (implicit).

2.4. Newton's Method

Since the reaction term is nonlinear in c_h^{n+1} , we apply Newton's method at each time step:

1. Set the initial guess $c_h^{n+1,(0)} = c_h^n$.
2. For $k = 0, 1, 2, \dots$ until convergence, solve for the increment $\delta_h \in V_h$:

$$J(c_h^{n+1,(k)})(\delta_h, v_h) = -R(c_h^{n+1,(k)})(v_h), \quad \forall v_h \in V_h, \quad (11)$$

where:

$$R(c)(v) = \int_{\Omega} \frac{c - c_h^n}{\Delta t} v \, dx + \theta a(c)(v) + (1 - \theta) a(c_h^n)(v),$$

$$J(c)(\delta, v) = \int_{\Omega} \frac{\delta}{\Delta t} v \, dx + \theta \int_{\Omega} D \nabla \delta \cdot \nabla v \, dx - \theta \int_{\Omega} \alpha (1 - 2c) \delta v \, dx.$$

3. Update

$$c_h^{n+1,(k+1)} = c_h^{n+1,(k)} + \delta_h,$$

and check $\|\delta_h\| < \varepsilon$ for convergence.

When converged, set $c_h^{n+1} = c_h^{n+1,(k+1)}$ and proceed to the next time step.

2.5. Algebraic Formulation

To solve the linearized problem (11) at each Newton iteration, we introduce $\{\varphi_i\}_{i=1}^{N_h}$, a basis of the finite element space V_h , where $N_h = \dim(V_h)$.

We can write now $\delta_h^{(k)}$ and $c_h^{n+1,(k)}$ as linear combinations of the basis functions:

$$\delta_h^{(k)}(\mathbf{x}) = \sum_{j=1}^{N_h} B_j^{(k)} \varphi_j(\mathbf{x}), \quad (12)$$

$$c_h^{n+1,(k)}(\mathbf{x}) = \sum_{j=1}^{N_h} C_j^{(k)} \varphi_j(\mathbf{x}), \quad (13)$$

where $\mathbf{B}^{(k)} = (B_1^{(k)}, B_2^{(k)}, \dots, B_{N_h}^{(k)})^T$ and $\mathbf{C}^{(k)} = (C_1^{(k)}, C_2^{(k)}, \dots, C_{N_h}^{(k)})^T$ are the coefficients of the linear combinations.

At each iteration of the Newton's method, we obtain the following linear system of equations:

$$\frac{1}{\Delta t} M \mathbf{B}^{(k)} + A^{n+1,(k)} \mathbf{B}^{(k)} = -\mathbf{R}^{n+1,(k)}, \quad (14)$$

where:

- $M \in \mathbb{R}^{N_h \times N_h}$ is the mass matrix and $M_{ij} = \int_{\Omega} \varphi_j \varphi_i \, dx$.
- $A^{n+1,(k)} \in \mathbb{R}^{N_h \times N_h}$ is the stiffness matrix at the k -th Newton iteration for the solution at time step $n+1$ and $A_{ij}^{n+1,(k)} = \theta J(c_h^{n+1,(k)})(\varphi_j, \varphi_i)$.
- $\mathbf{R}^{n+1,(k)} \in \mathbb{R}^{N_h}$ is the r.h.s. at the k -th Newton iteration for the solution at time step $n+1$ and $R_i^{n+1,(k)} = R(c_h^{n+1,(k)})(\varphi_i)$.

As we can see, the mass matrix does not depend on k and n , while, on the contrary, the stiffness matrix and the r.h.s. change at each Newton iteration and at each time step. For this reason, we have to compute them for each $c_h^{n+1,(k)}$ and all these computations are mutually independent, making them perfect for parallelization, as explained in Section 3.4.

2.6. Convergence Study

In order to analyze convergence of the Fisher-Kolmogorov equation, we need to consider both stability and accuracy.

It can be verified that the semi discrete formulation of the problem is stable under the assumptions introduced in Section 2. [3]

For the fully discrete formulation, stability depends on the choice of θ : it is granted for $\theta \geq \frac{1}{2}$, while for $\theta < \frac{1}{2}$ high values of Δt can generate instability. Forward Euler ($\theta = 0$) is conditionally stable, so it won't be considered in our implementation; on the other hand, Backward Euler ($\theta = 1$) and Crank-Nicolson ($\theta = \frac{1}{2}$) are unconditionally stable.

Accuracy in space depends on r , the degree of the finite elements:

$$\|c - c_h\|_{H^1(\Omega)} = O(h^r), \quad \|c - c_h\|_{L^2(\Omega)} = O(h^{r+1}). \quad (15)$$

Accuracy on time depends on the integration scheme we are using: Forward and Backward Euler are first order accurate, while Crank-Nicolson is a second order scheme.

$$\|c(t_n) - c_h^n\| = \begin{cases} O(\Delta t), & \text{Forward/Backward Euler (first order),} \\ O(\Delta t^2), & \text{Crank-Nicolson (second order).} \end{cases} \quad (16)$$

Putting space and time together, the global discretization error satisfies the estimate:

$$\|c(t_n) - c_h^n\|_{L^2(\Omega)} \leq C(h^{r+1} + \Delta t^p), \quad (17)$$

where r is the degree of the finite elements and p is the order of the chosen time integration scheme ($p = 1$ for Forward/Backward Euler or $p = 2$ for Crank-Nicolson).

Finite element degree r	$H^1(\Omega)$ error	$L^2(\Omega)$ error
$r = 1$ (linear)	$O(h^1)$	$O(h^2)$
$r = 2$ (quadratic)	$O(h^2)$	$O(h^3)$
$r = 3$ (cubic)	$O(h^3)$	$O(h^4)$

Table 2: Spatial convergence rates for finite elements of degree r .

Scheme	Stability	Order of accuracy
Forward Euler ($\theta = 0$)	Conditionally stable	$O(\Delta t)$
Backward Euler ($\theta = 1$)	Unconditionally stable	$O(\Delta t)$
Crank-Nicolson ($\theta = \frac{1}{2}$)	Unconditionally stable	$O(\Delta t^2)$

Table 3: Stability and temporal accuracy of the θ -schemes.

For our analysis, we chose to fix $r = 1$ and to use the Crank-Nicolson scheme, as stated in Section 3.3. As a consequence, we can rewrite (17) specifically for our work as:

$$\|c(t_n) - c_h^n\|_{L^2(\Omega)} \leq C(h^2 + \Delta t^2), \quad (18)$$

3. Solver Description

In the next section, we describe the numerical implementation of the above formulations using the `deal.II` library [4], with particular attention to the construction of the meshes, the choice of axonal approximations and the solution of the linear system.

3.1. Pre-Processing

In our study, we used both 2D and 3D meshes: the 2D mesh was fundamental to test and verify the correctness of the code, to regulate the parameters and to experiment quickly with elliptical axons (explained in greater detail in Section 3.2), since the equations in 2D are much simpler, while the 3D one gave us the occasion to extend our model to a realistic version.

In both cases, we split the white and gray matter in order to have a more accurate representation of the propagation of brain disease.

You can find all `.msh` files at [this link](#)

3.1.1 2D Sagittal Mesh

The 2D mesh was translated from `.dat` format with the python script `convert_mesh_dat_msh.py` in `mesh/scripts/`. The source.dat files were gently provided by Dr. Ellen Kuhl and are the same 2D mesh used in paper [1].

$d_{ext}^{GM} = d_{ext}^{WM}$	$d_{axn}^{GM} = d_{axn}^{WM}$	α_{GM}	α_{WM}
$5 \text{ mm}^2/\text{year}$	$50 \text{ mm}^2/\text{year}$	1 year^{-1}	2 year^{-1}

Table 4: Parameters chosen for 2D Sagittal mesh.

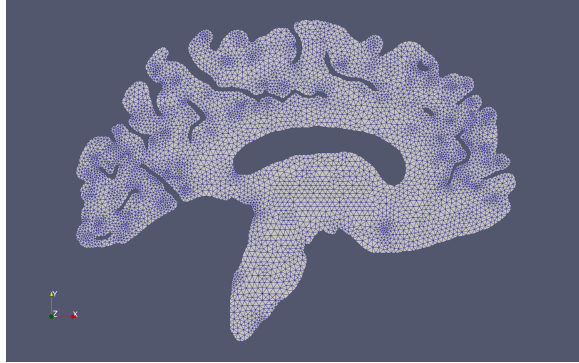


Figure 1: 2D mesh of the sagittal brain section with 6,441 points and 11,883 triangular elements.

The Python scripts also assign physical groups to mesh elements based on the White/Gray matter classification in the `.dat` files. Since the original classification was per-node instead of per-element, the element class is determined by majority-vote and the result is represented in Fig. 2.

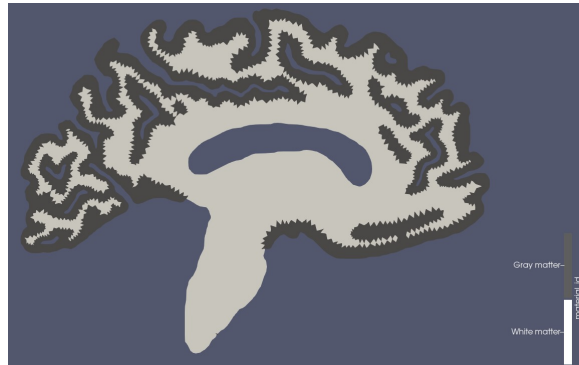


Figure 2: Distinction between white and gray matter in 2D Sagittal mesh.

3.1.2 3D "BrainCoarse" Mesh

After the 2D mesh, we focused our work on 3D meshes to better capture the complexity of the geometry of the brain.

The first 3D mesh used in our work was obtained from the BigBrain dataset [5] in `.nii` format, and subsequently converted into `.stl` and finally into `.msh` using Gmsh.

d_{ext}	d_{axn}	α
$3 \text{ mm}^3/\text{year}$	$40 \text{ mm}^3/\text{year}$	$1, 2 \text{ year}^{-1}$

Table 5: Parameters chosen for 3D Brain mesh.

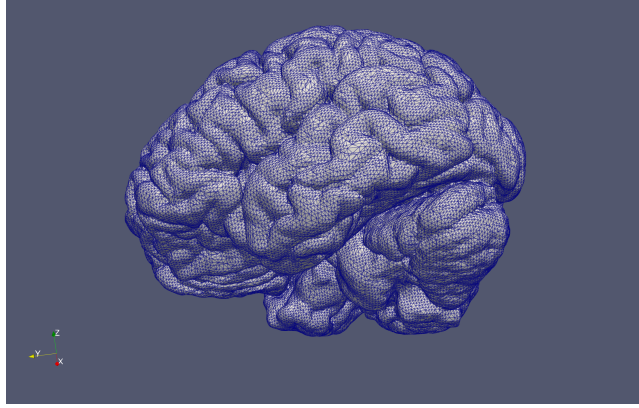


Figure 3: 3D Brain mesh consisting of 257,122 points, 1,338,867 tetrahedral volume elements and 172,780 triangular surface elements.

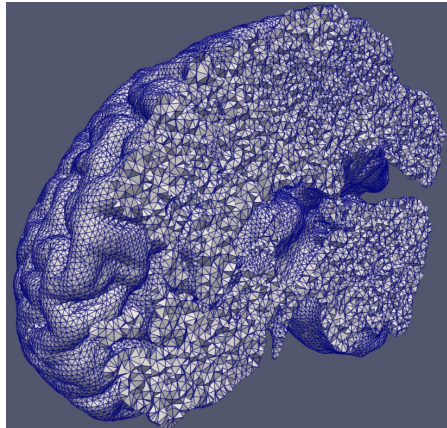


Figure 4: Section view of the 3D Brain mesh (Fig. 3), highlighting the internal tetrahedral structure.

This first version is a highly refined volumetric tetrahedral mesh, with over 1.3 million elements, ensuring accurate spatial resolution of the brain's complex geometry. However, we are still not able to capture some anatomical information, such as the distinction between white and gray matter, which could be important for a clear analysis of the development of neurodegenerative diseases. For this reason, we introduce two additional meshes: Ernie and MNI.

3.1.3 3D Ernie Mesh

We downloaded the data from the SimNIBS dataset [6], which provides a whole-head mesh. To isolate the brain and retain only the grey and white matter, we used the Python script `extract_ernie_brain.py` located in `mesh/scripts/ernie`.

$d_{ext}^{GM} = d_{ext}^{WM}$	$d_{axn}^{GM} = d_{axn}^{WM}$	α_{GM}	α_{WM}
$3 \text{ mm}^3/\text{year}$	$50 \text{ mm}^3/\text{year}$	0.6 year^{-1}	1.2 year^{-1}

Table 6: Parameters chosen for 3D Ernie mesh.

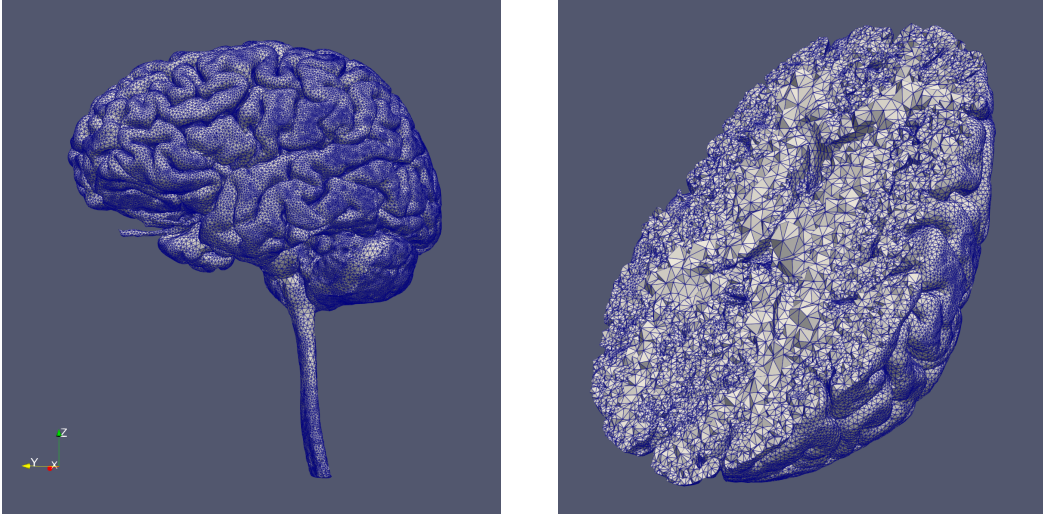


Figure 5: 3D Ernie mesh consisting of 400,148 points and 1,843,516 tetrahedral elements.



Figure 6: Distinction between white and gray matter in 3D Ernie mesh.

This mesh introduces the white/gray classification while still maintaining a large number of elements. However, we can still add some physical features such as cerebrospinal fluid (CSF) or ventricles to better represent the physical model.

3.1.4 3D MNI Mesh

This 3D mesh is the most advanced we have tested. It was translated from `.inp` format to `.msh` using a Python script and the Meshio [7] library.

The source `.inp` file was gently provided by Dr. Johannes Weickenmeier [1].

$d_{ext}^{GM} = d_{ext}^{WM}$	$d_{axn}^{GM} = d_{axn}^{WM}$	α_{GM}	α_{WM}
$3 \text{ mm}^3/\text{year}$	$50 \text{ mm}^3/\text{year}$	0.6 year^{-1}	1.8 year^{-1}

Table 7: Parameters chosen for the 3D MNI mesh. CSF and ventricles are not reported in the table since they have all coefficients equal to zero.

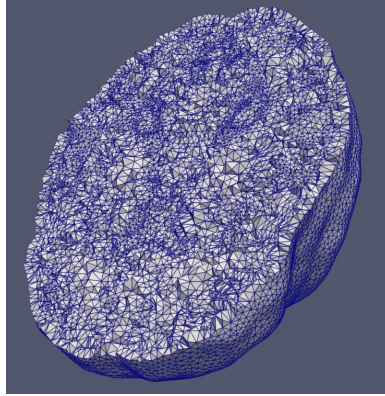


Figure 7: 3D MNI mesh consisting of 198,852 points and 1,123,575 tetrahedral elements.

Compared to the previous meshes, MNI classifies features as CSF, white/gray matter and ventricles, as shown in Fig. 8, providing us the most realistic representation of the human brain while maintaining a reasonable number of elements. So, it offers the best trade-off between anatomical detail and computational cost.



Figure 8: Distinction between white and gray matter in 3D MNI mesh.

3.2. Axon Approximation

We studied four models of brain structure: each of them represents fiber geometry and orientation in a different way, offering varying levels of biological accuracy. These four fibers orientations are the isotropic, the circumferential, the radial and the axon-based one, as shown in Fig. 9.

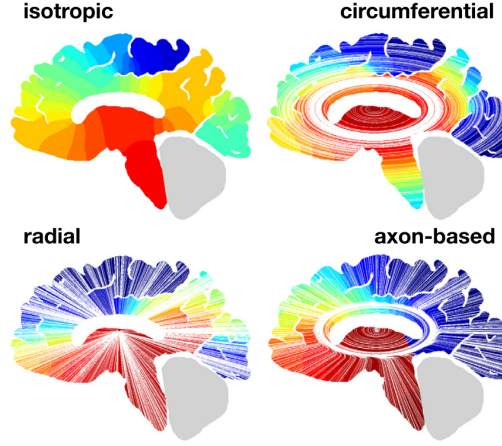


Figure 9: Fiber orientations in different configurations. [1]

Mathematically, the formulation of these models is given by:

$$D = d_{\text{ext}}I + d_{\text{axn}} \mathbf{n} \otimes \mathbf{n}, \quad (19)$$

as previously explained.

In the isotropic model, no preferred direction is present; in circumferential and radial models, $\mathbf{n}$ follows idealized tangential or radial patterns; in the axon-based model, $\mathbf{n}$ follows the actual direction, capturing realistic anisotropy in white matter.

In our work, the axon-based orientation was implemented by leveraging what we computed for the circumferential and the radial models: we defined a threshold to switch between the two, applying the circumferential orientation inside the threshold region and the radial orientation outside. A clear 2D representation is shown in Fig. 10, however the reasoning still applies in 3D cases.

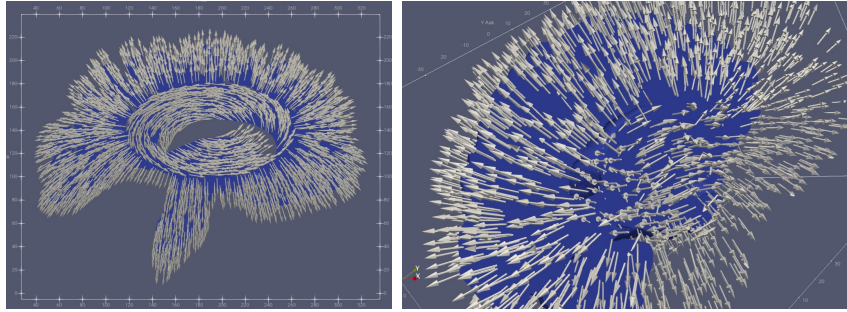


Figure 10: Approximation of axon-based using circumferential and radial.

3.3. Temporal Discretization Strategy

In our initial implementation, we used the Backward Euler scheme for time integration: we chose it because it is unconditionally stable, which makes it better than Forward Euler, conditionally stable. However, Backward Euler is a first order method and, as expected, we could observe an accumulation of the error and lack of accuracy with the progression of time.

To fix this problem, we replaced the time integration scheme with the Crank-Nicolson method. This method is second order accurate in time and still unconditionally stable for our problem: so, it is the best solution in terms of stability and accuracy.

3.4. Linear Solver and Domain Decomposition

At each Newton iteration we solve a large sparse linear system with a generally non-symmetric Jacobian. For this reason, we use GMRES: it is more robust than CG, which requires SPD matrices, and typically more reliable than BiCGStab.

As a preconditioner, SOR is used to accelerate convergence thanks to its simplicity and low computational cost.

Also, the linear systems at each time step are independent, as already stated in Section 2.5, so we exploit MPI-based domain decomposition to assemble and solve in parallel over subdomains.

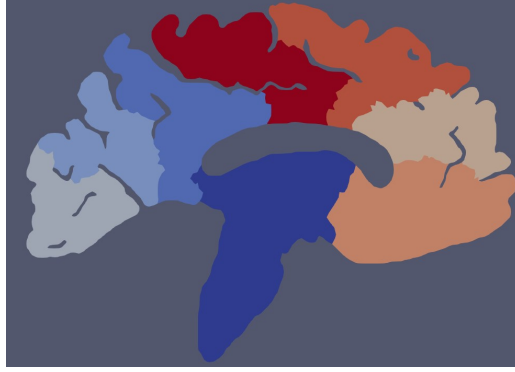


Figure 11: Domain decomposition of the 2D sagittal brain mesh in 8 subdomains.

3.5. Fixing the Negative Solution Problem

Let us recall that the solution of the Fisher-Kolmogorov equation is $c(\mathbf{x}, t) \in [0, 1]$: due to its physical meaning, the numerical scheme we constructed had to be positivity-preserving. [2]

From a theoretical point of view, in order to grant a positivity-preserving scheme, we must consider:

$$c = e^\lambda, \quad \lambda = \lambda(\mathbf{x}, t), \quad (20)$$

where λ is the new unknown of the problem.

The Fisher-Kolmogorov equation can be rewritten as follows:

Given a generic $t \in (0, T]$, $T > 0$ and $\mathbf{x} \in \Omega \subset \mathbb{R}^3$, find $\lambda = \lambda(x, t)$ such that:

$$\begin{cases} \frac{\partial e^\lambda}{\partial t} - \nabla \cdot (e^\lambda D \nabla \lambda) - \alpha e^\lambda (1 - e^\lambda) = 0, & \text{in } \Omega \times (0, T], \\ D \nabla \lambda \cdot \mathbf{n} = 0, & \text{on } \partial\Omega \times (0, T], \\ \lambda(t = 0) = \lambda_0, & \text{in } \Omega. \end{cases} \quad (21)$$

However, this formulation cannot be used in the implementation phase since it introduces a stronger nonlinearity and it is extremely computationally expensive. For this reason, in our implementation we preferred a simpler technique, the manual clamping: at each time step, after Newton's method converges, the solution is manually clamped to the interval $[0, 1]$ in order to ensure physical admissibility.

4. Numerical Results

In this section, we report the outcomes of the numerical simulations, focusing on the accuracy of the method, the qualitative behavior of the solution and the possible applicability to realistic brain geometries.

4.1. Numerical Validation

In order to validate the correctness of our implementation, we carried out a temporal convergence study, working on the 2D Sagittal mesh for simplicity. Since we had no analytical solution available, we picked a reference solution computed with a very small time step ($\Delta t = 1/192$ years ≈ 2 days), which is assumed to be sufficiently accurate. The numerical solutions obtained with larger time steps were then compared against this reference and the errors were evaluated in the L^2 and H^1 norms.

As shown in Fig. 12, the errors decrease with a rate proportional to $\mathcal{O}(\Delta t^2)$ in both norms, which confirms the second-order accuracy of the Crank–Nicolson time integration scheme and demonstrates that the solver reproduces the theoretical convergence properties of the method.

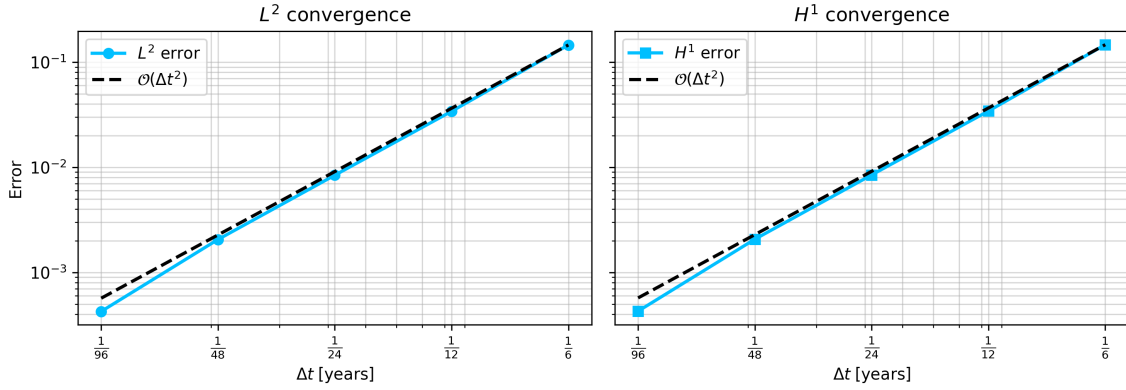


Figure 12: Temporal convergence of the Crank-Nicolson scheme. The errors are measured in L^2 (left) and H^1 (right) norms and they both exhibit a second-order decay $\mathcal{O}(\Delta t^2)$, confirming the theoretical accuracy of the scheme.

4.2. Sensitivity Analysis

We analyzed how the propagation is affected by different modeling choices and parameters. All the tests in this section are performed on the 2D Sagittal mesh, which is sufficient to capture the main behaviors while keeping the computational cost limited. The same analysis could be extended to 3D meshes, but this is not necessary if the goal is simply to compare the effect of different diffusion tensors and parameter variations.

4.2.1 Comparison of Diffusion Models

Firstly, we compared the effects of the four different fiber orientations explained in Section 3.2 to verify the importance of an axon-based model.

Figure 13 reports the temporal evolution of the solution for the isotropic, circumferential, radial and axon-based diffusion tensors. As expected, the isotropic model produces a symmetric spreading pattern around the initial point, which does not reflect the physical structure of the brain. Instead, the circumferential and radial models highlight the role of anisotropy but they remain strongly idealized. Finally, the axon-based model reproduces a realistic situation, mixing up the circumferential and the radial models.

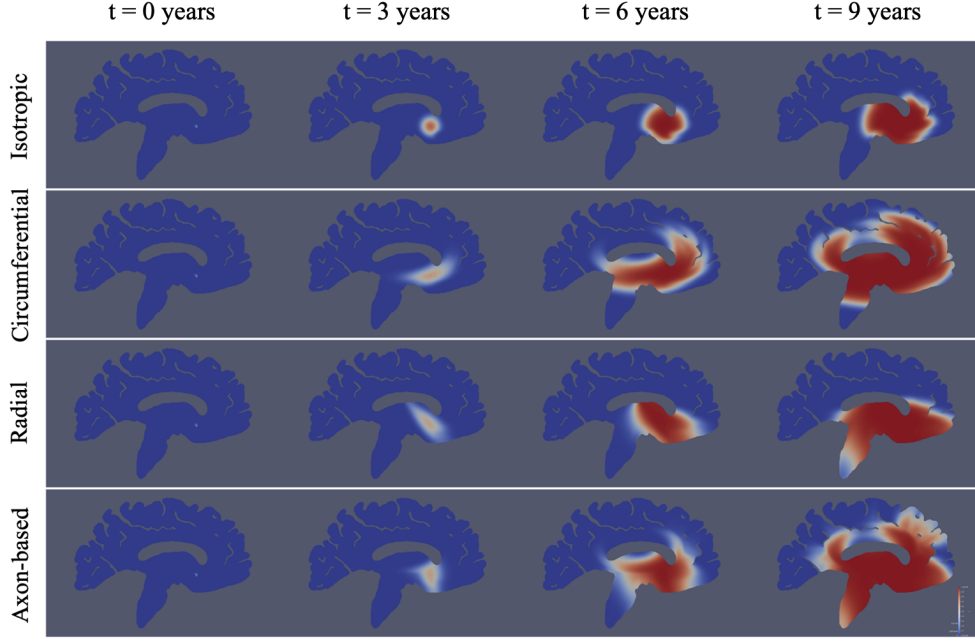


Figure 13: Temporal evolution of the disease for isotropic, circumferential, radial and axon-based diffusion models at different time points ($t = 0, 3, 6, 9$ years).

4.2.2 Parameters Variation

We then focused on the effect of the parameters on the propagation of the disease and in particular we studied what happens when modifying the extracellular diffusion coefficient d_{ext} , the axonal diffusion coefficient d_{axn} and the reaction rate α .

In the reference case, the diffusion coefficients are defined as:

$$d_{\text{ext}} = d_{\text{ext}}^{\text{GM}} = d_{\text{ext}}^{\text{WM}} = 5 \text{ [mm}^2\text{/year]}, \quad d_{\text{axn}} = d_{\text{axn}}^{\text{GM}} = d_{\text{axn}}^{\text{WM}} = 50 \text{ [mm}^2\text{/year]},$$

while the reaction rates are:

$$\alpha^{\text{GM}} = 1 \text{ [1/year]}, \quad \alpha^{\text{WM}} = 2 \text{ [1/year]}.$$

Starting from these values, we considered the following variations:

- d_{ext} : baseline, $4\times$, $8\times$,
- d_{axn} : baseline, $4\times$, $8\times$,
- α : $0.5\times$, baseline, $2\times$ (simultaneously in white and gray matter).

These perturbations allow us to evaluate the sensitivity of the model with respect to changes in diffusion and reaction parameters.

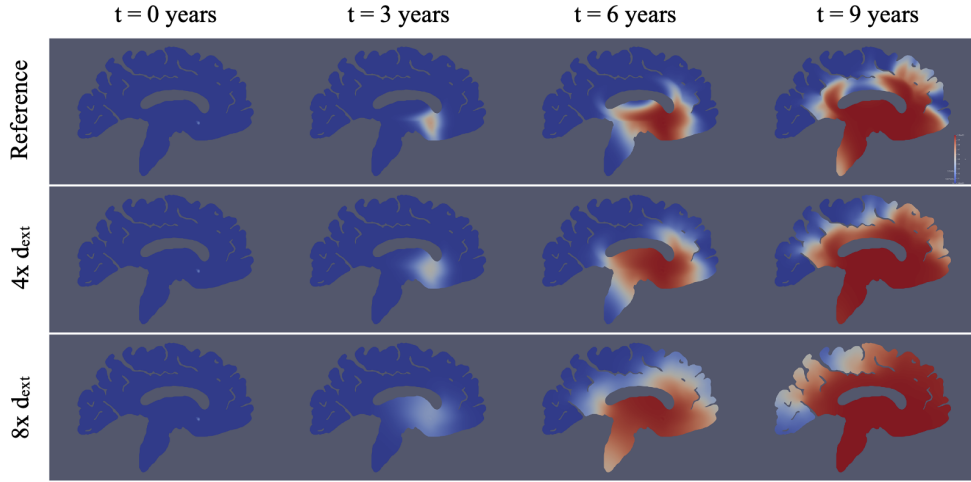


Figure 14: Temporal evolution of the disease for different extracellular diffusion coefficients: reference case, $4 \times d_{\text{ext}}$ and $8 \times d_{\text{ext}}$, at time points $t = 0, 3, 6, 9$ years.

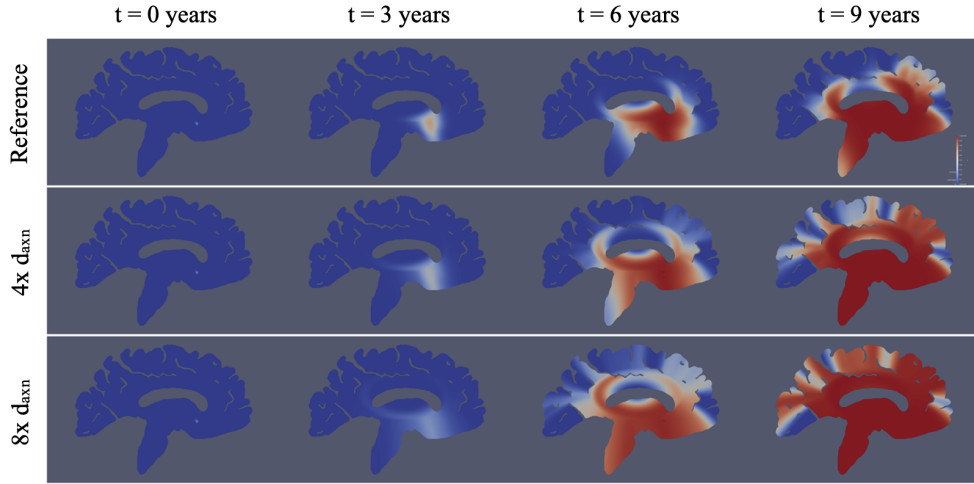


Figure 15: Temporal evolution of the disease for different axonal diffusion coefficients: reference case, $4 \times d_{\text{ext}}$ and $8 \times d_{\text{ext}}$, at time points $t = 0, 3, 6, 9$ years.

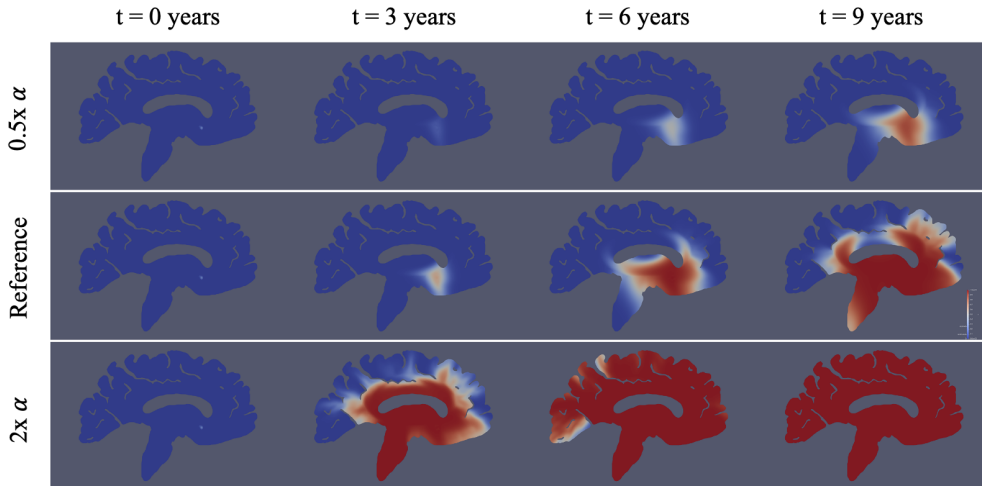


Figure 16: Temporal evolution of the disease for different reaction rates: $0.5 \times \alpha$, reference case, and $2 \times \alpha$, at time points $t = 0, 3, 6, 9$ years.

From our simulations, we can observe that both d_{ext} and d_{axn} promote a faster propagation when their values increase. However, the effect is qualitatively different: a higher d_{ext} enhances an expression more uniformly across the domain, while a higher d_{axn} channels the disease preferentially along the fiber orientations in an anisotropic way and reaching distant regions earlier.

In contrast, varying the reaction rate α mainly affects the temporal evolution of the growth: larger values accelerate the local conversion and lead to earlier saturation, but the spatial pattern remains essentially unchanged.

4.2.3 Variation of Seeding Region

To conclude our sensitivity analysis, we focused our tests on the role of the initial seeding region, as the choice of the starting point has a strong impact on the propagation of the disease.

As highlighted in [8], propagation tends to be faster along white matter tracts, while cortical gray matter is less permissive and leads to slower, more isotropic spreading.

In our simulations, this is reflected by the different patterns in Figure 17:

- the White Matter region (row 1) shows rapid propagation along anisotropic directions, typical of white matter tracts;
- the Near Ventricles region (row 2) shows a concentrated spread that is canalized and confined within the surrounding white matter;
- the Corpus Callosum region (row 3) shows a fast and wide expansion across the central white matter, acting as a bridge inside the brain;
- the Outer Cortex region (row 4) shows slower and more localized growth, consistent with the presence of gray matter.

Considering multiple seeding points therefore allows us to capture the variability of disease progression depending on its anatomical origin.

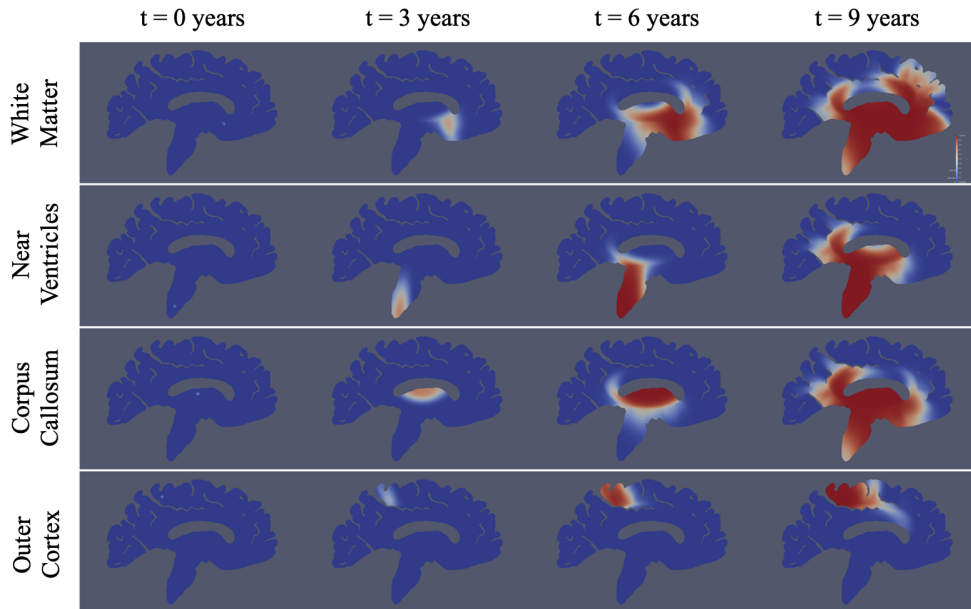


Figure 17: Temporal evolution of the disease with different initial seeding regions at time points $t = 0, 3, 6, 9$ years. Each row corresponds to a distinct initial location.

4.3. Simulations on 3D Meshes

Similar results are obtained on the 3D meshes but we report below only the MNI mesh for the sake of brevity.

The initial seeding region was set near the center of the brain for ease of visualization and to better highlight the differences in material that MNI allows.

Material	d_{ext}	d_{axn}	α
Gray Matter	3 mm ³ /year	50 mm ³ /year	0.6 year ⁻¹
White Matter	3 mm ³ /year	50 mm ³ /year	1.8 year ⁻¹
CSF	0 mm ³ /year	0 mm ³ /year	0 year ⁻¹
Ventricles	0 mm ³ /year	0 mm ³ /year	0 year ⁻¹

Table 8: Parameters chosen for this simulation on the 3D MNI mesh.

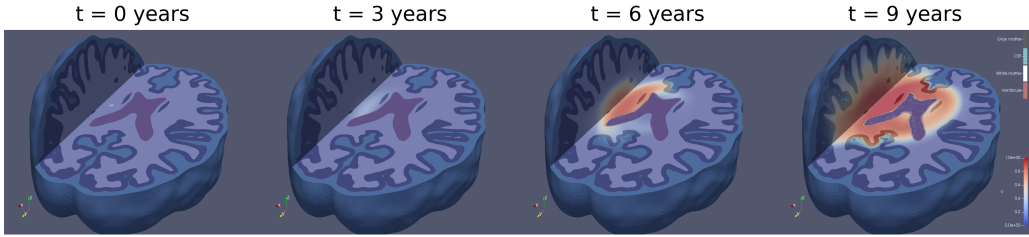


Figure 18: Temporal evolution of the disease in the 3D MNI mesh at time points $t = 0, 3, 6, 9$ years.

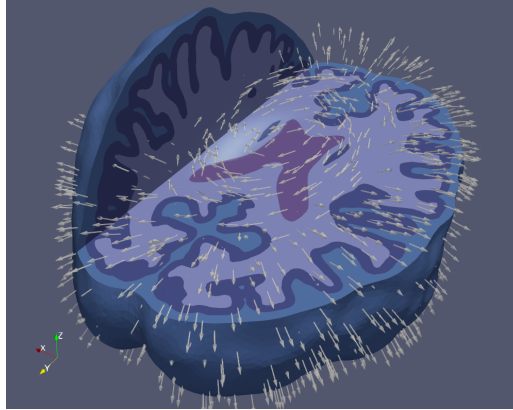


Figure 19: Axons of the MNI mesh, approximated using circumferential and radial, as explained above in Figure 10.

You can find full animations of our results at [this link](#).

4.4. Parallel Performance

To evaluate the parallel performance of our MPI implementation, we conducted a scaling analysis using a set of brain mesh presets. The benchmarking process was automated via a custom Bash script, and ran on our laptop (16 cores, 16 GB memory).

It first validates each mesh preset, then proceeds with timed runs. For every preset, we performed sequential executions as a baseline and then repeated the simulations in parallel using MPI, spanning a range of process counts (2, 4, 8, 12, 16). Each configuration was executed multiple times to mitigate random fluctuations, with elapsed times recorded and aggregated in a CSV file for post-processing.

This methodology allowed us to capture both the speedup achieved by increasing the number of parallel processes (strong scaling), as well as the efficiency when scaling up the problem size proportionally to the number of processes (weak scaling).

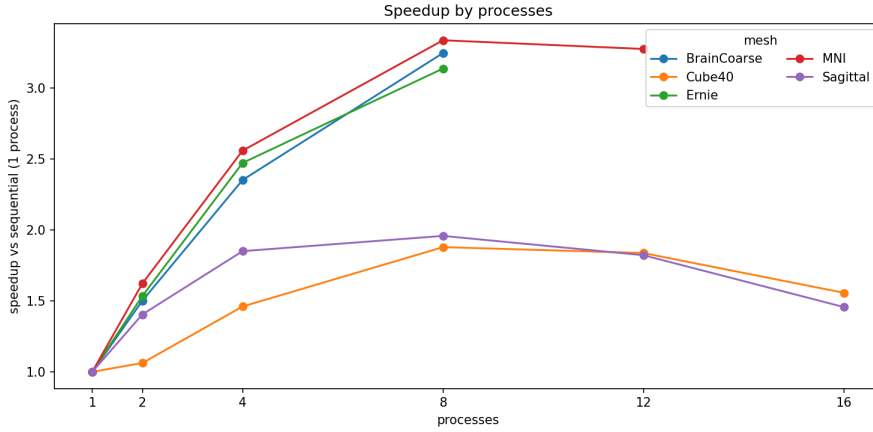


Figure 20: Performance evaluation: speedup across different meshes.

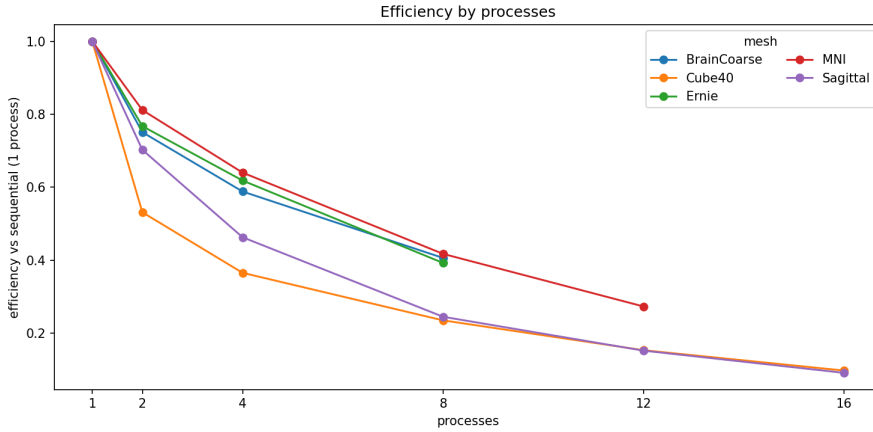


Figure 21: Efficiency across different meshes and thread counts.

The speedup is close to linear up to 4-8 processes on the larger meshes, while smaller meshes saturate sooner due to the lower workload compared to communication overhead.

Beyond 8-12 processes this workload-to-overhead ratio gets too small for the larger meshes as well.

At 12-16 processes we also hit memory limits because each MPI rank loads its own copy of the mesh. Despite this, the overall trend is still visible. The practical sweet spot on this hardware is around 8 processes.

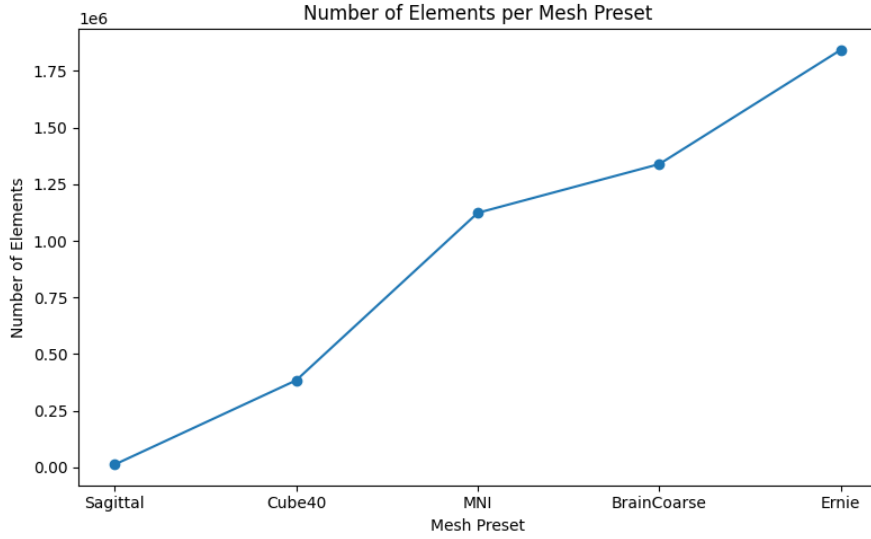


Figure 22: Mesh sizes, based on number of cells.

Find all our speedup analysis results at [this link](#).

5. Conclusions

In this work, we developed and tested a finite element solver for the Fisher-Kolmogorov equation, using the `deal.II` library and exploiting MPI-based domain decomposition for parallel efficiency.

We progressively validated our approach on meshes of increasing complexity, starting from a simplified 2D sagittal section and moving to realistic 3D brain geometries. The introduction of white/gray matter differentiation, cerebrospinal fluid and ventricles allowed us to assign heterogeneous parameters, resulting in more biologically consistent simulations.

The numerical experiments confirmed the accuracy of the Crank-Nicolson scheme and highlighted the different roles of isotropic and axonal diffusion, as well as the influence of the reaction coefficient. The axon-based tensor was crucial to reproduce realistic propagation patterns, channeling the disease preferentially along white matter tracts.

Parallel performance analysis showed that the code scales well with the number of MPI processes, thanks to distributed mesh handling and Trilinos solvers. Nonetheless, further improvements could be achieved by employing more advanced preconditioners and reducing the cost of output operations.

Overall, our study demonstrates that combining advanced numerical schemes, parallelization techniques and anatomically informed meshes provides an effective framework to investigate neurodegenerative disease progression.

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